



Agilent N6705B DC power analyser

Automotive communication test systems

Electronic control units communicate with each other using some protocol like controller area network (CAN), local interconnect network (LIN), FlexRay, and media-oriented systems transport (MOST) for multimedia and infotainment.

"Debugging and testing this type of network topology can be challenging and time-consuming," says Sun. Pathak adds, "Due to huge in-vehicle communication and stringent emission norms, it necessitates data logging for a long duration along with the ability to replay the same."

The popular usage of CAN in automobiles has also prompted the development of a variety of CAN interfacing tools. However, Gautam says, "after the design phase is over, a recursive approach is generally followed by using test and measurement instruments and going back to conduct modifications based on the actual test results until perfect values are physically achieved. No prototype can afford to proceed for further stages of design cycle by completely substituting actual measurements with simulation."

Jain says, "To test these serial data protocols and hence the communication between various ECUs, an oscilloscope with CAN or LIN decoding and triggering capability is used. Also, one can use a logic analyser to perform timing and state analysis on the

bitstreams and command execution."

ECU: critical to test

The electronic control unit (ECU) is the heart of automotive electronics. It takes care of the engine control, transmission, airbags, antilock braking system, audio system, windows, doors, mirror adjustments, environment control and many other such systems and sub-systems. "It is difficult to prioritise the automotive assemblies in terms of their relative importance because the basic operation of an automobile is completely dependent on coherent operation of all the modules, big or small," says Gautam.

"Using ECUs to control both essential and non-essential automotive systems is becoming the new industry standard. Based on digital technology, these ECUs require a deeper understanding of complex timing and signal integrity issues," shares Sun.

Testing of automotive electronics relies heavily on being able to interact with a particular ECU responsible for a particular electronic sub-system. Panikkar says, "Initially, these test systems are used to test prototypes of the ECU in a rapid-control prototyping scenario, where they provide real-world signals to the prototype. They are used for fault insertion and studying the behaviour of the sub-system under varying conditions and limits." With vehicles incorporating more and more electronics, testing of driver information systems, navigation as well as in-vehicle entertainment systems is also

necessary and critical.

EMI/EMC: a rising concern

The staggering growth of electronics in automobiles has led to a rising concern about electromagnetic compatibility (EMC) and electromagnetic interference (EMI) issues. Telematics, electric power, use of electronic controls in place of mechanical, in-vehicle infotainment, wireless voice and digital communication, active suspension, antilock brake and satellite navigation are some of the functional units that give rise to EMI/EMC issues. T&M vendors are developing new methods, tools and standards for automotive EMC testing.

"For EMI/EMC testing, you require EMI receivers. Apart from this, T&M providers provide antennae, amplifiers, signal generators, power meters and other test transducers required for automotive EMC testing," says Gautam.

RF testing

Use of radio frequency (RF) interfaces within an automobile for communication, safety and security systems has created the need to develop and measure real-time RF systems that can efficiently monitor vehicle operation and status.

"Tyre pressure monitoring, keyless entry, infotainment and immobiliser systems are examples of technologies containing RF signals that are often triggered by external events which create profiles that are intermittent in nature—present one moment, absent the next—and variable over time," informs Sun. These transient signals are difficult to measure using traditional spectrum analysis tools and require new instruments that can measure time-varying signals in the frequency domain.

"For RF testing, the test instruments used are vector/scalar signal generators, analysers and vector network analysers generally less than 6 GHz," says Gautam. There are also handheld RF analysers with spectrum analyser capability which can be used to fulfil RF testing needs of the automotive industry.